MIL-H-13514F
30 March 1988
SUPERSEDING
MIL-H-13514E
29 June 1983

MILITARY SPECIFICATION

HEATERS, SPACE, RADIANT TYPE, PORTABLE 45,000 BTU/HR.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers 45,000 BTU/HR, portable radiant type space heaters (see 6.1).

1.2 Classification. Heaters shall be of the following types, as specified (see 6.2):

Type I - Coal or solid fuel burning
Type II - Oil burning

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

	FF-P-386	- Pins, Cotter (Split)
	QQ-W-461	- Wire, Steel, Carbon, (Round, Bare and Coated)
*	PPP-B-601	- Boxes, Wood, Cleated-Plywood
*	PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

	MIL-P-116	- Preservation, Methods of
	MIL-P-551	- Pipes and Elbows, Air Conditioning - Heating
	MIL-P-584	- Poker
	MIL-D-1427	- Damper, Flue, 4-Inch
	MIL-B-2029	- Burner, Assembly, Space Heater
	MIL-C-10872	- Cap, Flue, 4-Inch, Cylindrical
	MIL-A-54003	- Arrester, Spark, M-1941

STANDARDS

MILITARY

	MIL-STD-105	- Sampling procedures and Tables for Inspection by Attributes
	MIL-STD-129	- Marking for Shipment and Storage
	MIL-STD-130	- Identification Marking of U.S. Military Property
*	MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Methods

* (Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

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DRAWINGS

US ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-11-82	-	Heater, Space, Radiant Type, Portable, 45,000 BTU/Hr.; Stove Assembly and Details
5-11-83	-	Heater, Space, Radiant Type, Portable, 45,000 BTU/Hr.; Details "A"
5-11-84	-	Heater, Space, Radiant Type, Portable, 45,000 BTU/Hr.; Details "B"
5-11-85	-	Heater, Space, Radiant Type, Portable, 45,000 BTU/Hr.; Details "C"

* (Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

	A 48	-	Gray Iron Castings
	A 109	-	Steel, Strip, Carbon, Cold-Rolled
*	A 622	-	Steel, Sheet and Strip, Carbon, Drawing Quality, Special Killed, Hot-Rolled
	D 3951	-	Standard Practice for Commercial Packaging

* (Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

* (Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

* 3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials and components. The materials and components shall be as specified herein and on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this document (see 6.4).

3.2.1 Steel.

3.2.1.1 Strip and sheet. Strip and sheet steel shall conform to ASTM A109 and A622 respectively.

3.2.1.2 Wire. Steel wire used for rivets shall conform to AISI 1010, finish 1, annealed of QQ-W-461.

3.2.2 Iron castings. Base metal for gray iron castings shall consist of not less than 40 percent pig iron and the remainder scrap iron. The castings shall conform to ASTM A48.

* 3.2.3 Pipe, air conditioning - heating, straight. The 4-inch diameter, 24-inch long, straight, air conditioning-heating pipe shall be a black, rust inhibiting paint finish and conform to MIL-P-551, class I. Six lengths of pipe shall be furnished with each heater.

* 3.2.4 Damper, flue (for Type I). The flue damper shall conform to MIL-D-1427.

* 3.2.5 Poker, 24 inch (for Type I). The 24 inch poker shall conform to MIL-P-584.

* 3.2.6 Pin, cotter (for Type I). The cotter pin shown on Drawing 5-11-85 shall conform to type A of FF-P-386.

* 3.2.7 Shovel, fire, hand (for Type I). The shovel, fire, hand shall be furnished as Government furnished property (see 6.2 and 6.2.1).

* 3.2.8 Arrester, spark (for Type I). The spark arrester shall conform to MIL-A-54003.

* 3.2.9 Cap, flue, 4-inch, cylindrical (for Type II). The 4-inch cylindrical cap flue shall conform to MIL-C-10872.

* 3.2.10 Burner, assembly (for Type II). The oil burner assembly shall conform to MIL-B-2029.

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3.3 Design and construction. The design and construction of the type I and type II heaters shall conform to the drawings as follows, and as specified herein:

a. Type I - Applicable drawings, 5-11-82, 5-11-83, 5-11-84, and 5-11-85
b. Type II - Applicable drawings, 5-11-82, except grates are not required, 5-11-83 and 5-11-84

3.3.1 Type I heater. The type I heater shall consist of the components in table I. The casting of the grate assembly shall be clean and free of warpage, blow holes, cold shut and cracks. The surface shall have a uniform appearance with no visible evidence of projections and metal break out from the mold. The draw grate when assembled shall be capable of being pushed in and out with no interference.

TABLE I. Components of type I heater

Quantity	Nomenclature
1	Base assembly
1	Grate assembly, round and draw
1	Adapter ring
1	Top assembly with lid assembly
6	Pipe, air conditioning - heating, straight, 4-inch diameter, 24-inches long, nested
1	Damper, flue
1	Poker, 24 inch
1	Shaker
1	Shovel, fire, hand
1	Arrester, spark

3.3.2 Type II heaters. The type II heater shall consist of the components in table II.

TABLE II. Components of type II heater

Quantity	Nomenclature
1	Base assembly
1	Adapter ring
1	Top assembly with lid assembly
6	Pipe, air-conditioning-heating, straight, 4-inch diameter, 24 inches long, nested
1	Burner, assembly, space heater
1	Cap, flue, 4-inch, cylindrical

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3.4 Marking for identification. Heaters shall be marked for identification in accordance with the unit marking requirements of MIL-STD-130 and as specified on applicable drawings.

* 3.5 Workmanship. Space heaters shall be clean and free of distortion and misalignment of parts. Spot welds shall be adherent and not burned through. All parts formed from sheet or strip steel shall be free of warpage, dents, wrinkles, bends not specified, fractures, sharp edges, burrs, and slivers. Casting and molded parts shall be free of sand, fins, pits, blow holes, and sprues. External surfaces shall be free from sharp edges and corners except where corners are required. The heaters shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.4.2 End item visual examination. The end items shall be examined for the defects listed in table III. The lot size shall be expressed in units of heaters. The sample unit shall be one heater. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE III. End item visual defects

Examine	Defect	Classification Major	Minor
Construction and workmanship	Accessory or component missing, separated, or malformed	X	
	Sheet metal fractured, dented, torn, wrinkled, or with open laminations	X	
	Sharp edges, burrs, or slivers		X
	Parts misaligned	X	
	Castings with prominent parting lines due to improper molding	X	
	Cracks	X	
	Blowholes, hard spots, cold shuts, or porosity	X	
	Not clean		X
	Projections or metal breakout from mold	X	
	Draw grate movement restricted due to interference	X	

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Top assembly	Collar improperly formed or trimmed, top edge of uneven height, sharp or turned outward, or loose	X	
	Spot welds broken, missing, or cratered due to metal burning	X	
	Heater top does not seat into adapter ring	X	
	Lid lug bent, affecting use		X
	Lid distorted or opened around flanged edge		X
	Lid latch not free moving or turning		X
	Spot welds on handle less than two	X	
Adapter ring	Adapter ring does not seat properly in heater base, or distorted	X	
	Top edge of flange uneven		X
Base assembly	Door does not fit contour of front of base (opening between door and base)	X	
*	Door latch or door loose	X	
*	Hinge not in proper location	X	
	Door latch embossing missing or too small		X
	Door latch not free turning		X
	Door latch rivet improperly peened or not peened		X
	Door out of alignment		X
	Hinge binds		X
Grates	Warped more than 1/8 inch in any direction	X	
Marking for identification	Missing, illegible, incomplete, wrong size, or wrong location	X	

* 4.4.3 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on drawing 5-11-82. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of heaters. The sample unit shall be one heater. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.0.

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* 4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
* Preservation	Improperly applied or missing
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container.
Content	Number per container is more or less than required.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservative application.

* 5.1.1.1.1 Type I and type II heaters. All uncoated ferrous metal surfaces of each heater and components, except the pipe sections shall be cleaned by method C-1, thoroughly dried and coated with type P-1 preservative in accordance with MIL-P-116. The pipe sections shall be coated with preservative in accordance with MIL-P-551. All components not otherwise specified shall be preserved in accordance with method III of MIL-P-116.

5.1.1.2 Unit packing.

* 5.1.1.2.1 Pipe sections. Six sections of pipe, prepared as specified in 5.1.1.1.1, shall be unit packed in accordance with MIL-P-551, except that prior to wrapping the pipe sections for the type I heater, one poker, one shaker, and one damper, prepared as specified in 5.1.1.1.1, shall be placed in the center of the pipe sections. The size of the fiberboard box shall not exceed outside dimensions of 25 by 6 by 6 inches.

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5.1.2 Commercial. The pipe sections, poker, shaker, and damper shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing.

* 5.2.1.1 Type I heater. One complete type I heater shall be packed into a close-fitting cleated-plywood box conforming to style A or I of PPP-B-601. The inside dimensions of the shipping container shall be approximately 21 by 21 by 27 inches. The heater components shall be arranged in the shipping container with the heater base being placed first and moved to one corner of the 21 by 21 inch surface. The heater top with lid secured, shall be inverted and placed inside the heater base with some cushioning material placed between the two components and adequate enough to support the weight of the heater. The fiberboard box specified in 5.1.1.2.1, shall be slid into the opposing corner to which the heater base was positioned. The grate assembly shall be wrapped with cushioning material and placed with the concave side down inside the heater top. The adapter ring shall be placed on top of the heater top assembly. The remaining components of the type I heater shall be packed in the available space in the shipping container. All components shall be cushioned, blocked, braced, and rendered immobile within the shipping container, in accordance with MIL-STD-1186.

* 5.2.1.2 Type II heater. One complete type II heater shall be packed by the same requirements as specified in 5.2.1.1, except that the grate assembly, damper flue, poker, shaker, shovel, and spark arrester will not be included. The burner assembly shall be packed within the inverted heater top assembly. All components shall be cushioned, blocked, braced, and rendered immobile within the shipping container, in accordance with MIL-STD-1186.

* 5.2.2 Level B packing.

* 5.2.2.1 Type I heater. One complete type I heater shall be packed as specified in 5.2.1.1, except the shipping container shall conform to a grade V13c, class weather-resistant, corrugated fiberboard container in accordance with PPP-B-636. The container shall be closed and reinforced in accordance with the appendix of PPP-B-636.

* 5.2.2.2 Type II heater. One complete type II heater shall be packed as specified in 5.2.1.2, except the shipping container shall conform to a grade V13c, class weather-resistant, corrugated fiberboard container in accordance with the appendix of PPP-B-636. The container shall be closed and reinforced in accordance with the appendix of PPP-B-636.

* 5.2.3 Commercial packing. Type I and II complete heaters, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

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* 5.3 Marking. In addition to any special markings required by the contract, interior packages and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951 as applicable.

6. NOTES

6.1 Intended use. The space heaters are intended to heat tents and temporary shelters.

6.1.1 Type I. The type I space heater is intended for use with coal or other solid fuel and may be converted to an oil burning space heater by removing the grate assembly and inserting the oil burner covered by MIL-B-2029.

6.1.2 Type II. The type II space heater is furnished with the oil burner covered by MIL-B-2029.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type space heater required (see 1.2).
c. When a first article is required (see 3.1).
d. Source and delivery of Government furnished property (see 3.2.7 and 6.2.1).
e. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

* 6.2.1 Government furnished property. The shovel should be obtained under NSN 5120-00-293-0450 (see 6.2).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

* 6.5 Subject term (keyword) listing.

Coal burning
Heater, space, radiant, portable
Heater, 45000 Btu/hr
Oil burning
Solid fuel burning

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* 6.6 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the previous issue.

Custodians:

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Air Force - 99

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DLA - CS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-13514F	Heaters, Space, Radiant Type, Portable 45,000 Btu/hr

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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